

DAVI/NFC Integration Product Backlog

A Core-Team-friendly delivery plan for NFC card assignment, Axon-owned check-in, freebie collection, kiosk operations, security, offline recovery, and event-day readiness.

CORE ARCHITECTURE DECISION

DAVI provides the NFC kiosk foundation. Axon owns the live operational truth.

Check-in and freebies must continue through Axon even if DAVI is unavailable during the event.

Axon is adding an NFC operating layer, not replacing its core platform.

Axon keeps ownership of admission, freebies, and fallback while NFC improves the event-day experience.

8

delivery epics

49

product stories

250

planning points

18

blocked until discovery
evidence arrives

For nontechnical teams

The card is only the key. Axon still checks if the attendee is allowed, whether the item was already claimed, and what to do if the scan fails.

For engineering

Build event-scoped APIs, immutable ledgers, idempotent retries, atomic duplicate prevention, kiosk credentials, and minimum offline cache.

For operations

Approve devices, train staff, rehearse exceptions, prepare QR fallback, assign owners, monitor stations, and reconcile after the event.

THIS BACKLOG IS ALIGNED TO THE READINESS CENTER MODEL

Every event-day capability must have an owner, a release gate, a rehearsal path, a visible blocker state, and stakeholder-safe reporting.

This mirrors the Core Team package: event-centric ownership, RACI-style accountability, readiness scoring, stakeholder visibility, and closure/reconciliation after the event.

What Axon already does today, and what NFC changes.

This anchors the NFC plan to Axon's real operating baseline.

Working now in Axon

- Event creation, public registration pages, and ticket selection already exist in Axon.
- Manual payment proof upload and verification workflows already exist for paid registrations.
- QR ticketing, attendee lookup, and live QR/manual check-in already exist in current operations.
- Attendance analytics, exports, and administrative reporting already exist as the operating baseline.

Required before NFC launch

- No NFC card registry, assignment history, replacement lifecycle, or card-to-attendee mapping exists yet.
- No kiosk/station credential model exists yet for event-scoped NFC devices.
- Current attendance records need a stronger immutable event ledger for NFC, retries, and overrides.
- No controlled offline NFC queue or freebie entitlement/collection workflow exists yet.

PLATFORM IMPACT

The NFC project extends Axon's existing registration, verification, QR, and attendance foundation into event-day operations.

In practical terms, Axon moves from being the place where attendees register and get validated to also being the place where card identity, entrance control, freebies, and reconciliation are governed.

How the organizer workflow changes with DAVI/NFC.

This page shows exactly where the NFC layer changes real event operations for organizers.

1

Create the event

CURRENT AXON STATE

Organizers already create events, tiers, and registration flows in Axon.

2

Collect registrations

CURRENT AXON STATE

Attendees already register, pay, upload proof if needed, and receive Axon ticket records.

3

Prepare event operations

NEW NFC IMPACT

Organizers will add approved devices, stations, card assignment, freebie rules, and fallback owners.

4

Run entrance and freebies

NEW NFC IMPACT

Attendees tap NFC cards, but Axon remains the authority for entry, stock release, duplicates, and exceptions.

5

Monitor and support

NEW NFC IMPACT

Leads watch station health, switch to QR/manual if needed, and resolve card or connectivity exceptions.

6

Close and reconcile

NEW NFC IMPACT

Axon reconciles attendance, freebies, offline queues, overrides, and post-event learnings.

What impact this brings to Axon Tickets.

The gain is not only speed at the kiosk. It is stronger event-day control inside the existing Axon platform.

Faster front-door flow

Entrance teams can move from scan-or-search to tap-and-decide while keeping QR/manual fallback.

Stronger operational control

Freebies, overrides, and duplicate prevention become governed inside Axon instead of manual side processes.

Better stakeholder visibility

Attendance and redemption become easier to explain in operations reviews, reconciliations, and future dashboards.

Higher platform value

Axon moves closer to an event operating system, not only a registration and ticketing system.

IMPORTANT OPERATING TRADEOFF

NFC can make the event feel smoother, but only after device certification, repository validation, staff training, and fallback rehearsal. The platform benefit comes from controlled rollout, not from skipping proof.

How this fits the Event Workspace and Readiness Center.

The NFC iteration becomes an event-readiness workstream with named owners, clear status, and rehearsed fallback.

Event Owner

Approves go/no-go, scope box, fallback policy, and stakeholder updates.

Event Manager

Coordinates readiness tasks, staffing, training, and rehearsal completion.

Registration Lead

Owns attendee eligibility, QR fallback, card assignment support, and check-in rules.

Freebie Lead

Owns entitlement rules, stock control, handover confirmation, and reconciliation.

Kiosk Tech Lead

Owns DAVI repository adaptation, approved devices, station setup, and diagnostics.

Security Owner

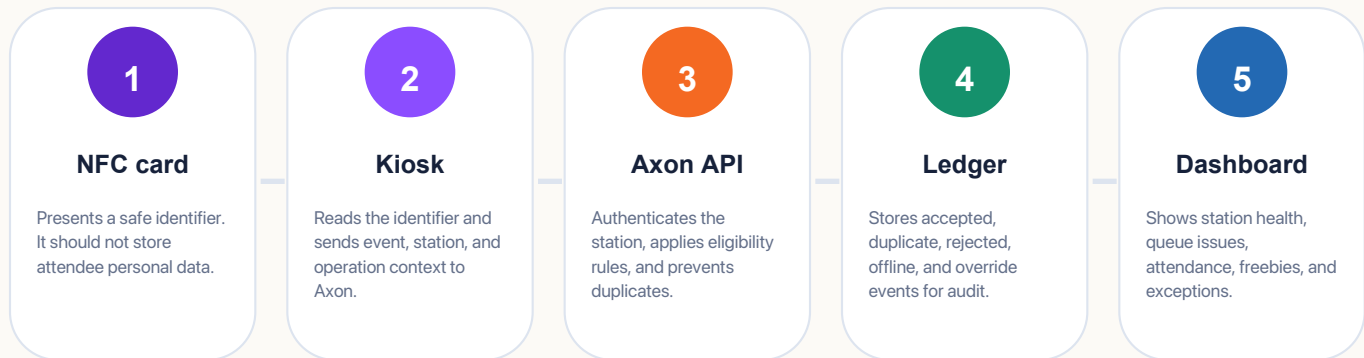
Owns kiosk credentials, local data protection, revocation, auditability, and privacy.

CORE TEAM DECISION TO MAKE EARLY

Confirm who has authority to switch from NFC to QR/manual baseline on event day.

This avoids hallway decision-making when queues are forming. The incident commander should be named before rehearsal.

The tap journey: simple for staff, strict behind the scenes.



What DAVI should not do in the live path

DAVI should not approve check-in, decide freebie eligibility, store the event attendance truth, or become the dependency that blocks the queue.

What Axon must own

Card assignment, attendee resolution, station authorization, duplicate prevention, audit logs, offline recovery, QR/manual fallback, dashboards, and reconciliation.

PHONE NFC NOTE

NFC exists on modern iPhones and Android phones, but implementation differs. Android browser Web NFC support cannot be assumed on iPhone Safari. If iPhones are production kiosks, validate native Core NFC or supported reader behavior before committing.

Scope, status, and delivery shape.

This is the working delivery package after the business context and organizer-workflow review.

45

must-have stories

6

discovery stories

39

July MVP stories

4

post-July stories

STATUS MIX



READY 31

BLOCKED 18

EPIC SUMMARY

Epic	Focus	Stories	Points	Blocked
DAVI-00	Discovery, Repository and Device Validation	6	21	4
DAVI-01	NFC Card Identity and Lifecycle	5	19	5
DAVI-02	Kiosk Identity, Authorization and Security	5	21	1
DAVI-03	Axon-Authoritative NFC Check-In	7	41	1
DAVI-04	Axon Freebie Entitlements and Collection	7	34	1
DAVI-05	Kiosk Application Adaptation and UX	5	24	3
DAVI-06	Limited Offline Mode and Synchronization	6	44	1
DAVI-07	Testing, Rehearsal, Release and Event Operations	8	46	2

Do not launch NFC because it exists. Launch it because it is proven.

G1

Repository

Required evidence: The repository DAVI sent for the NFC kiosk builds and its external dependencies are documented.

If failed: Stop NFC work and escalate.

G2

Card/device

Required evidence: Real cards read reliably on every approved production phone/reader.

If failed: Reduce device scope or QR-only.

G3

Data/security

Required evidence: Card lifecycle, station permissions, and offline dataset are approved.

If failed: No production credential rollout.

G4

Correctness

Required evidence: Atomic check-in and freebie concurrency tests pass.

If failed: No multi-station launch.

G5

Operations

Required evidence: Full rehearsal, fallback, monitoring, and runbook are complete.

If failed: Run QR/manual baseline only.

DEFINITION OF DONE FOR JULY STORIES

- Acceptance criteria pass with real event rules.
- Plain-language kiosk messages are reviewed by operations.
- Authorization, idempotency, retries, and duplicate prevention are tested.
- Existing QR/manual flows still work.
- Runbook, owners, monitoring, and rollback path are ready.

The safest build order.

1

Discovery

Build the DAVI-sent kiosk repository, identify card protocol, test devices, freeze business rules.

2

Foundations

Card records, station permissions, credentials, immutable ledgers.

3

Core flows

NFC check-in, freebie validation, handover confirmation, fallback.

4

Kiosk adaptation

Replace live DAVI dependencies with Axon endpoints and plain states.

5

Offline and recovery

Minimum encrypted cache, provisional queue, idempotent sync.

6

Rehearsal and release

Device tests, concurrency tests, runbook, incident command, reconciliation.

GREATEST-VALUE NEXT STORY

DAVI-DISC-01 - Receive and build the kiosk repository

DAVI already sent a repository we can use as the starting point for the NFC kiosk. Building and inspecting it first unlocks card behavior, dependency mapping, device decisions, realistic estimates, and safe API design.

DAVI-00 - Discovery, Repository and Device Validation

Prove that the contributed kiosk technology and real NFC cards work on the exact July devices before Axon commits to implementation.

EPIC OUTCOME

Prove that the contributed kiosk technology and real NFC cards work on the exact July devices before Axon commits to implementation.

DAVI-DISC-01

Receive and build the kiosk repository

MUST

BLOCKED

3 SP

DISCOVERY

Plain English: the team understands the DAVI-sent repository that can be used as the starting point for the NFC kiosk.

User need: As a technical lead, I need to receive, build, and inspect the repository DAVI sent for the NFC kiosk.

Acceptance checks

- Given repository access and setup instructions,, when the documented build is run,, then the kiosk launches without undocumented manual patches.
- Given the running application,, when network activity is inspected,, then every DAVI/external dependency is documented.
- Given build failure,, when discovery ends,, then the blocker, owner and required DAVI action are recorded.

Depends on: Repository access from DAVI/Dotside

Note: DAVI has already sent a repository that can be used for the NFC kiosk. This first story proves how much of it is reusable, what it depends on, and what Axon must replace or own.

DAVI-DISC-02

Identify the card protocol and identifier

MUST

BLOCKED

5 SP

DISCOVERY

Plain English: Axon can design a stable and secure card mapping.

User need: As a integration architect, I need to determine exactly what the kiosk reads from each card.

Acceptance checks

- Given a production card,, when scanned repeatedly,, then the returned identifier and protocol are captured and consistent.
- Given multiple cards,, when scanned,, then identifiers are unique at the expected collision risk.
- Given UID, NDEF or proprietary data,, when identified,, then cloning, rewriting and privacy implications are documented.

Depends on: DAVI-DISC-01, Production NFC cards

DAVI-00 continued

Remaining user stories for this epic.

DAVI-DISC-03

Approve the July kiosk device matrix

MUST

BLOCKED

5 SP

DISCOVERY

Plain English: production does not depend on assumed NFC support.

User need: As a event technology lead, I need to test the exact phone, operating system and app/browser combinations.

Acceptance checks

- Given each proposed device,, when the real card is scanned 20 consecutive times,, then success/failure rate and median read time are recorded.
- Given a browser-based Web NFC kiosk,, when tested,, then Android Chrome, HTTPS, foreground and permission requirements are confirmed.
- Given iPhone use,, when proposed,, then a native Core NFC implementation or supported reader is demonstrated; Safari Web NFC is not assumed.

Depends on: DAVI-DISC-01, DAVI-DISC-02

DAVI-DISC-04

Document the existing kiosk architecture

MUST

BLOCKED

3 SP

DISCOVERY

Plain English: the adaptation work is estimable.

User need: As a developer, I need to map screens, local storage, APIs, authentication and offline behavior.

Acceptance checks

- Given the repository,, when reviewed,, then a component/data-flow diagram is produced.
- Given local persistence,, when present,, then stored fields, encryption and expiry are documented.
- Given DAVI production calls,, when found,, then each is classified as remove, replace, retain outside live operations, or needs decision.

Depends on: DAVI-DISC-01

DAVI-00 continued

Remaining user stories for this epic.

DAVI-DISC-05

Freeze check-in and freebie business rules

MUST

READY

3 SP

DISCOVERY

Plain English: developers implement one unambiguous behavior.

User need: As a product owner, I need to approve eligibility, duplicate, replacement and override rules.

Acceptance checks

- Given registration states,, when rules are approved,, then every state has an explicit check-in decision.
- Given each July freebie,, when rules are approved,, then eligible tiers, quantity, variants, reversal and override behavior are documented.
- Given re-entry or repeat taps,, when rules are approved,, then the intended result and displayed message are defined.

DAVI-DISC-06

Approve the July scope box and release gate

MUST

READY

2 SP

DISCOVERY

Plain English: scope remains achievable and reliable.

User need: As a product owner, I need to separate must-have pilot work from post-July enhancements.

Acceptance checks

- Given the discovery results,, when scope is approved,, then NFC check-in, fallback, security and one controlled freebie pilot are explicitly included or excluded.
- Given any new request,, when raised after scope freeze,, then it replaces an item of equivalent effort or moves after July.
- Given a failed discovery gate,, when reviewed,, then QR-only operation remains the approved fallback.

Depends on: DAVI-DISC-01, DAVI-DISC-02, DAVI-DISC-03

DAVI-01 - NFC Card Identity and Lifecycle

Axon can safely link a physical card to an attendee while preserving assignment history, replacement and revocation.

EPIC OUTCOME

Axon can safely link a physical card to an attendee while preserving assignment history, replacement and revocation.

DAVI-CARD-01 Design NFC card records

MUST

BLOCKED

3 SP

JULY MVP

Plain English: the physical credential can be managed without storing PII on the card.

User need: As a system owner, I need to store a safe card token and lifecycle state.

Acceptance checks

- Given a card,, when registered,, then Axon stores an internal ID, token/token hash, protocol/type, status and timestamps.
- Given the database design,, when reviewed,, then no attendee name, email, payment details or entitlement list is required on the NFC card.
- Given sensitive raw card values,, when stored,, then hashing/encryption requirements are explicitly decided.

Depends on: DAVI-DISC-02

DAVI-CARD-02 Design attendee-card assignment history

MUST

BLOCKED

5 SP

JULY MVP

Plain English: lost and reassigned cards remain explainable.

User need: As a operations administrator, I need to link cards to attendees without losing replacement history.

Acceptance checks

- Given a card assignment,, when created,, then attendee, event scope, assigner, time and active state are recorded.
- Given one active card,, when assigned to another attendee,, then Axon blocks silent reassignment.
- Given an attendee with an active card,, when a replacement is issued,, then the previous assignment is deactivated and retained.

Depends on: DAVI-CARD-01

DAVI-01 continued

Remaining user stories for this epic.

DAVI-CARD-03 Assign a card to a verified attendee

MUST

BLOCKED

5 SP

JULY MVP

Plain English: the card can identify the correct person at the event.

User need: As a authorized event staff, I need to search an attendee and tap a card to assign it.

Acceptance checks

- Given a verified attendee without an active card,, when staff confirms a scanned unused card,, then assignment succeeds.
- Given an unverified attendee,, when assignment is attempted,, then it is rejected unless the approved rule explicitly permits pre-assignment.
- Given success or rejection,, when completed,, then the actor, station, reason and time are audited.

Depends on: DAVI-CARD-02, DAVI-STN-03

DAVI-CARD-04 Replace a lost or damaged card

MUST

BLOCKED

3 SP

JULY MVP

Plain English: the attendee can continue without duplicate active credentials.

User need: As a authorized support staff, I need to revoke the old card and issue a replacement.

Acceptance checks

- Given a reported lost card,, when replacement is confirmed,, then the old card becomes revoked before the new card becomes active.
- Given the old card,, when later tapped,, then Axon returns card revoked and does not perform the event action.
- Given replacement,, when complete,, then the reason and staff identity are retained.

Depends on: DAVI-CARD-03

DAVI-01 continued

Remaining user stories for this epic.

DAVI-CARD-05

Search and reconcile card assignments

SHOULD

BLOCKED

3 SP

AFTER JULY

Plain English: unlinked and duplicate operational cases can be resolved.

User need: As a event support lead, I need to find cards by attendee, token reference or assignment state.

Acceptance checks

- Given an attendee or safe card reference,, when searched,, then authorized staff can view current and prior assignments.
- Given unassigned or revoked cards,, when filtered,, then the support team can export an exception list.

Depends on: DAVI-CARD-02

DAVI-02 - Kiosk Identity, Authorization and Security

Kiosk devices receive only the permissions needed for their assigned event and station.

EPIC OUTCOME

Kiosk devices receive only the permissions needed for their assigned event and station.

DAVI-STN-01

Model event stations

MUST

READY

3 SP

JULY MVP

Plain English: every operation has a known physical and technical origin.

User need: As a event administrator, I need to register entrance, freebie and support stations.

Acceptance checks

- Given a station,, when created,, then event, station type, display name, device reference and active state are stored.
- Given a station type,, when permissions are evaluated,, then only approved actions are allowed.

DAVI-STN-02

Issue short-lived kiosk credentials

MUST

READY

5 SP

JULY MVP

Plain English: a copied or lost credential has limited use.

User need: As a security owner, I need to authenticate each registered kiosk independently.

Acceptance checks

- Given an approved device,, when activated,, then Axon issues event- and station-scoped credentials.
- Given an expired or revoked credential,, when an operation is submitted,, then Axon rejects it.
- Given source code review,, when completed,, then production secrets are not hard-coded in the kiosk repository.

Depends on: DAVI-STN-01

DAVI-02 continued

Remaining user stories for this epic.

DAVI-STN-03

Enforce least-privilege station permissions

MUST

READY

5 SP

JULY MVP

Plain English: a kiosk cannot access full administration.

User need: As a platform owner, I need to restrict stations to check-in, assignment or freebie actions.

Acceptance checks

- Given an entrance station,, when it requests user management or event deletion,, then access is forbidden.
- Given a freebie station,, when it attempts card assignment without permission,, then access is forbidden.
- Given permissions,, when enforced,, then checks occur on the API, not only in the interface.

Depends on: DAVI-STN-02

DAVI-STN-04

Remotely revoke a kiosk

MUST

READY

3 SP

JULY MVP

Plain English: event operations can contain security incidents.

User need: As a incident commander, I need to disable a lost or compromised device.

Acceptance checks

- Given an active station,, when revoked,, then new online operations are rejected immediately.
- Given queued offline actions from a revoked device,, when synchronized,, then they are quarantined for review.

Depends on: DAVI-STN-02

DAVI-02 continued

Remaining user stories for this epic.

DAVI-STN-05

Protect local kiosk data

MUST

BLOCKED

5 SP

JULY MVP

Plain English: device loss does not expose unnecessary attendee data.

User need: As a security owner, I need to encrypt and expire cached event information.

Acceptance checks

- Given locally cached attendee/card data,, when stored,, then it is encrypted using platform-appropriate secure storage.
- Given event completion or cache expiry,, when cleanup runs,, then local data and credentials are removed.
- Given the minimum dataset,, when reviewed,, then payment proofs, bank details, auth secrets and unnecessary PII are absent.

Depends on: DAVI-DISC-04, DAVI-OFF-01

DAVI-03 - Axon-Authoritative NFC Check-In

A card tap becomes a fast Axon check-in while preserving the existing QR and manual pathways.

EPIC OUTCOME

A card tap becomes a fast Axon check-in while preserving the existing QR and manual pathways.

DAVI-CHK-01

Create an immutable check-in event model

MUST

READY

5 SP

JULY MVP

Plain English: attendance decisions remain auditable.

User need: As a operations owner, I need to record every accepted, duplicate, rejected and overridden attempt.

Acceptance checks

- Given a check-in attempt,, when processed,, then attendee, event, method, result, station, actor/device, operation ID and timestamps are stored.
- Given a correction or override,, when processed,, then the original event remains unchanged and a new event explains the change.

DAVI-CHK-02

Make accepted check-in atomic

MUST

READY

8 SP

JULY MVP

Plain English: simultaneous stations cannot create conflicting attendance.

User need: As a event operator, I need to allow only one accepted check-in per attendee and event.

Acceptance checks

- Given two concurrent first attempts,, when committed,, then exactly one is accepted.
- Given the second attempt,, when returned,, then it reports already checked in with the authoritative time.
- Given retries with one operation ID,, when repeated,, then the original result is returned.

Depends on: DAVI-CHK-01

DAVI-03 continued

Remaining user stories for this epic.

DAVI-CHK-03

Resolve an NFC card to an eligible attendee

MUST

BLOCKED

5 SP

JULY MVP

Plain English: Axon can apply its registration and ticket rules.

User need: As a entrance kiosk, I need to submit a card token and selected event.

Acceptance checks

- Given an active assigned card,, when submitted,, then Axon resolves the attendee without returning unnecessary PII.
- Given unknown, revoked, replaced or wrong-event cards,, when submitted,, then Axon returns a clear business result.
- Given a DAVI outage,, when the Axon API is available,, then the online transaction still completes.

Depends on: DAVI-CARD-02, DAVI-STN-03

DAVI-CHK-04

Apply existing Axon eligibility rules

MUST

READY

5 SP

JULY MVP

Plain English: NFC does not weaken admission controls.

User need: As a entrance staff, I need to validate event, payment and ticket state after a card tap.

Acceptance checks

- Given verified registration or paid valid ticket according to approved rules,, when tapped,, then check-in may proceed.
- Given cancelled, rejected, refunded or wrong-event state,, when tapped,, then Axon rejects the action.
- Given rule results,, when returned,, then the kiosk displays a plain-language message.

Depends on: DAVI-DISC-05, DAVI-CHK-03

DAVI-03 continued

Remaining user stories for this epic.

DAVI-CHK-05

Expose the NFC check-in API

MUST

READY

5 SP

JULY MVP

Plain English: the kiosk can safely retry and receive a deterministic result.

User need: As a kiosk application, I need to submit event, card, station and unique operation context.

Acceptance checks

- Given a valid station request,, when POST /event-operations/check-ins/nfc is called,, then Axon returns checked_in, already_checked_in or a documented rejection code.
- Given the same operation ID,, when retried,, then no additional event is created.
- Given invalid station scope,, when submitted,, then access is forbidden and logged.

Depends on: DAVI-CHK-02, DAVI-CHK-03, DAVI-CHK-04

DAVI-CHK-06

Preserve QR and manual fallback

MUST

READY

5 SP

JULY MVP

Plain English: hardware or card failure does not stop the event.

User need: As a event staff, I need to switch from NFC to existing admission methods.

Acceptance checks

- Given an unreadable or unlinked card,, when support is required,, then staff can scan the existing Axon QR or search the attendee.
- Given any method,, when attendance succeeds,, then the same authoritative check-in ledger is used.
- Given a prior NFC check-in,, when QR is scanned,, then the duplicate result is shown.

Depends on: DAVI-CHK-01, DAVI-CHK-02

DAVI-03 continued

Remaining user stories for this epic.

DAVI-CHK-07

Check-in operations dashboard

SHOULD

READY

8 SP

AFTER JULY

Plain English: issues can be handled before queues grow.

User need: As a event lead, I need to monitor station health, check-ins, duplicates and exceptions.

Acceptance checks

- Given active stations,, when dashboard data refreshes,, then last activity, success count, rejection count and offline queue state are visible.
- Given an exception spike,, when thresholds are exceeded,, then the operational lead can identify the station and reason.

Depends on: DAVI-CHK-01, DAVI-STN-01

DAVI-04 - Axon Freebie Entitlements and Collection

Axon determines who may receive each physical item and prevents duplicate collection.

EPIC OUTCOME

Axon determines who may receive each physical item and prevents duplicate collection.

DAVI-FRB-01

Define event freebies

MUST

READY

3 SP

JULY MVP

Plain English: the station knows what may be released.

User need: As a event administrator, I need to configure a freebie, stock and optional variants.

Acceptance checks

- Given an event,, when a freebie is created,, then name, description, active period, stock quantity and optional variants are stored.
- Given an inactive freebie,, when scanned,, then no new collection is accepted.

DAVI-FRB-02

Assign attendee entitlements

MUST

READY

5 SP

JULY MVP

Plain English: eligibility is consistent and reviewable.

User need: As a event administrator, I need to grant freebies by event, tier or explicit attendee.

Acceptance checks

- Given an approved tier rule,, when entitlements are generated,, then eligible attendees receive the configured quantity.
- Given an explicit exception,, when added,, then the source and actor are recorded.
- Given no entitlement,, when validation occurs,, then the result is not eligible.

Depends on: DAVI-FRB-01, DAVI-DISC-05

DAVI-04 continued

Remaining user stories for this epic.

DAVI-FRB-03

Validate a freebie before handover

MUST

BLOCKED

5 SP

JULY MVP

Plain English: staff can make the correct decision before releasing stock.

User need: As a freebie station, I need to tap a card and see eligibility and prior collection.

Acceptance checks

- Given an active card and entitlement,, when validated,, then the kiosk shows item, variant, allowance and prior claim state.
- Given an ineligible or unknown card,, when validated,, then no collection is created.

Depends on: DAVI-CARD-02, DAVI-FRB-02, DAVI-STN-03

DAVI-FRB-04

Require staff confirmation of physical handover

MUST

READY

3 SP

JULY MVP

Plain English: a mere card tap does not consume stock.

User need: As a freebie staff member, I need to confirm that the item was actually given.

Acceptance checks

- Given a successful validation,, when the screen appears,, then collection is not recorded until staff confirms handover.
- Given cancellation or timeout,, when no confirmation occurs,, then the entitlement remains unused.

Depends on: DAVI-FRB-03

DAVI-04 continued

Remaining user stories for this epic.

DAVI-FRB-05

Prevent duplicate freebie collection atomically

MUST

READY

8 SP

JULY MVP

Plain English: two stations cannot release the same limited item twice.

User need: As a event owner, I need to allow only the entitled quantity.

Acceptance checks

- Given quantity one,, when two stations confirm simultaneously,, then exactly one redemption is accepted.
- Given the losing or repeated request,, when returned,, then the original collection time and station are shown.
- Given one operation ID,, when retried,, then the original result is returned.

Depends on: DAVI-FRB-04

DAVI-FRB-06

Controlled override and reversal

SHOULD

READY

5 SP

AFTER JULY

Plain English: unusual cases can be handled without destroying auditability.

User need: As a authorized supervisor, I need to record an exception with a reason.

Acceptance checks

- Given an ineligible or previously collected result,, when override is attempted,, then supervisor permission and a mandatory reason are required.
- Given a reversal,, when permitted,, then a new reversal event is written and the original redemption remains visible.

Depends on: DAVI-FRB-05

DAVI-04 continued

Remaining user stories for this epic.

DAVI-FRB-07

Reconcile digital redemptions with stock

MUST

READY

5 SP

JULY MVP

Plain English: loss and data discrepancies are visible.

User need: As a freebie lead, I need to compare accepted claims, overrides and remaining physical inventory.

Acceptance checks

- Given a freebie,, when reconciliation runs,, then starting stock, accepted redemptions, reversals, overrides and expected remainder are shown.
- Given a discrepancy,, when identified,, then it can be assigned and annotated without changing original records.

Depends on: DAVI-FRB-05

DAVI-05 - Kiosk Application Adaptation and UX

The contributed kiosk becomes a clear Axon-operated interface that event staff can use under pressure.

EPIC OUTCOME

The contributed kiosk becomes a clear Axon-operated interface that event staff can use under pressure.

DAVI-KSK-01

Replace live DAVI transaction dependencies

MUST

BLOCKED

8 SP

JULY MVP

Plain English: DAVI availability does not block event operations.

User need: As a kiosk developer, I need to route check-in and freebie decisions directly to Axon.

Acceptance checks

- Given repository API calls,, when adaptation is complete,, then check-in and freebie calls use Axon endpoints.
- Given non-operational DAVI community features,, when retained,, then they are isolated from admission and collection success.

Depends on: DAVI-DISC-04, DAVI-CHK-05, DAVI-FRB-03

DAVI-KSK-02

Create explicit kiosk states

MUST

BLOCKED

5 SP

JULY MVP

Plain English: I always know what the station is doing.

User need: As a event staff, I need to see ready, reading, validating, success and exception states.

Acceptance checks

- Given the kiosk,, when idle,, then a prominent ready-to-tap instruction is shown.
- Given a tap,, when processing,, then duplicate input is temporarily locked and progress is visible.
- Given each Axon business result,, when returned,, then a distinct color, title, instruction and optional attendee name are displayed.

Depends on: DAVI-DISC-01

DAVI-05 continued

Remaining user stories for this epic.

DAVI-KSK-03

Show plain-language errors and fallback

MUST

READY

3 SP

JULY MVP

Plain English: I can resolve the queue without engineering knowledge.

User need: As a event staff, I need to receive actionable messages rather than technical codes.

Acceptance checks

- Given card_not_found,, when returned,, then the kiosk instructs staff to search or scan QR.
- Given wrong_event or not_verified,, when returned,, then the reason and support action are clear.
- Given timeout,, when displayed,, then retry and fallback options are available without creating duplicates.

Depends on: DAVI-CHK-05

DAVI-KSK-04

Display connectivity and station identity

MUST

READY

3 SP

JULY MVP

Plain English: misconfiguration is caught before it affects attendees.

User need: As a event lead, I need to see whether the kiosk is online and correctly assigned.

Acceptance checks

- Given the application header,, when active,, then event, station and connectivity state are visible.
- Given wrong event/station configuration,, when detected,, then operations are blocked until corrected.

Depends on: DAVI-STN-01, DAVI-STN-02

DAVI-05 continued

Remaining user stories for this epic.

DAVI-KSK-05

Accessible and device-tested kiosk interface

MUST

BLOCKED

5 SP

JULY MVP

Plain English: the physical workflow remains fast and inclusive.

User need: As a event staff, I need to operate the kiosk reliably on the approved device fleet.

Acceptance checks

- Given approved kiosk devices,, when used in portrait/landscape as designed,, then controls remain legible and touch targets are at least 44 pixels.
- Given success or rejection,, when displayed,, then color is not the only indicator.
- Given sunlight/noisy venue conditions,, when rehearsed,, then visual and optional audio/haptic feedback remain understandable.

Depends on: DAVI-DISC-03, DAVI-KSK-02

DAVI-06 - Limited Offline Mode and Synchronization

A controlled station can continue provisionally during connectivity loss without pretending disconnected devices have perfect global duplicate prevention.

EPIC OUTCOME

A controlled station can continue provisionally during connectivity loss without pretending disconnected devices have perfect global duplicate prevention.

DAVI-OFF-01 Define the minimum encrypted offline dataset

MUST

READY

5 SP

JULY MVP

Plain English: offline capability does not become uncontrolled PII replication.

User need: As a security and operations owner, I need to cache only the event data needed for provisional decisions.

Acceptance checks

- Given the cache design,, when approved,, then it includes event, safe card lookup, attendee ID/display name, eligibility, entitlement and last-known use state only as needed.
- Given excluded data,, when reviewed,, then payment proof, bank details, auth secrets, fraud metadata and unnecessary contact details are absent.
- Given a cache version,, when expired,, then the kiosk stops using it.

DAVI-OFF-02 Record provisional offline check-ins

MUST

READY

8 SP

JULY MVP

Plain English: one entrance can continue during an outage.

User need: As a designated offline entrance, I need to validate cached cards and queue signed operations.

Acceptance checks

- Given a valid cached attendee,, when Axon is unreachable,, then the designated kiosk records a provisional operation with unique ID and local sequence.
- Given a repeat on the same device,, when attempted,, then it is shown as already provisionally checked in.
- Given an unresolvable card,, when offline,, then the kiosk routes to manual supervisor handling.

Depends on: DAVI-OFF-01, DAVI-CHK-01

DAVI-06 continued

Remaining user stories for this epic.

DAVI-OFF-03

Persist the offline queue safely

MUST

BLOCKED

5 SP

JULY MVP

Plain English: provisional operations are not silently lost.

User need: As a kiosk operator, I need to retain queued actions through app restart or battery loss.

Acceptance checks

- Given queued operations,, when the app restarts,, then unacknowledged records remain.
- Given local storage,, when inspected,, then records are encrypted and credentials are stored using secure platform facilities.
- Given acknowledged synchronization,, when complete,, then retention follows the approved cleanup policy.

Depends on: DAVI-DISC-04, DAVI-STN-05

DAVI-OFF-04

Synchronize idempotently after reconnection

MUST

READY

8 SP

JULY MVP

Plain English: recovery is reliable.

User need: As a kiosk application, I need to upload queued operations without duplicates.

Acceptance checks

- Given reconnection,, when queued operations are submitted,, then Axon validates device, event, operation ID and sequence.
- Given the same batch is submitted twice,, when processed,, then original results are returned.
- Given a conflict,, when detected,, then it is retained for reconciliation rather than silently overwritten.

Depends on: DAVI-OFF-02, DAVI-OFF-03

DAVI-06 continued

Remaining user stories for this epic.

DAVI-OFF-05

Restrict offline freebie operation

MUST

READY

5 SP

JULY MVP

Plain English: physical stock is not released twice across disconnected stations.

User need: As a freebie lead, I need to use only one offline station per freebie type.

Acceptance checks

- Given connectivity loss,, when offline freebie mode is approved,, then only the designated station may create provisional claims.
- Given another station,, when it attempts offline collection for that item,, then it is blocked.
- Given offline handover,, when confirmed,, then a visible manual stock tally and supervisor identity are required.

Depends on: DAVI-FRB-05, DAVI-OFF-01

DAVI-OFF-06

Multi-station local venue coordination

COULD

READY

13 SP

AFTER JULY

Plain English: offline duplicate prevention approaches online reliability.

User need: As a event technology lead, I need to coordinate disconnected stations over a local venue server.

Acceptance checks

- Given a local venue network,, when stations are disconnected from the internet but connected locally,, then accepted actions are shared through one authority.
- Given local-server failure,, when it occurs,, then the system falls back to the approved single-station procedure.

Depends on: DAVI-OFF-04

DAVI-07 - Testing, Rehearsal, Release and Event Operations

The NFC pilot is proven with real hardware, realistic concurrency, trained staff and an explicit fallback before attendees arrive.

EPIC OUTCOME

The NFC pilot is proven with real hardware, realistic concurrency, trained staff and an explicit fallback before attendees arrive.

DAVI-QA-01

Automate card, check-in and freebie API tests

MUST

READY

8 SP

JULY MVP

Plain English: core rules remain stable during iteration.

User need: As a development team, I need to cover success, rejection, idempotency and authorization.

Acceptance checks

- Given unit/integration tests,, when run,, then active, unknown, revoked, wrong-event and ineligible card cases are covered.
- Given repeated operation IDs,, when tested,, then no duplicate accepted action is created.
- Given unauthorized station scope,, when tested,, then the API returns forbidden.

DAVI-QA-02

Run concurrency and retry tests

MUST

READY

5 SP

JULY MVP

Plain English: duplicate prevention is proven rather than assumed.

User need: As a quality engineer, I need to simulate simultaneous taps and network timeouts.

Acceptance checks

- Given at least 20 concurrent attempts for one attendee,, when executed,, then one accepted check-in exists.
- Given concurrent freebie confirmation,, when executed,, then accepted quantity never exceeds entitlement.
- Given timeout followed by retry,, when tested,, then idempotency returns the original result.

Depends on: [DAVI-CHK-02](#), [DAVI-FRB-05](#)

DAVI-07 continued

Remaining user stories for this epic.

DAVI-QA-03

Test the approved device and card matrix

MUST

BLOCKED

5 SP

JULY MVP

Plain English: hardware surprises are removed.

User need: As a event technology lead, I need to validate every production phone/reader and card combination.

Acceptance checks

- Given each approved combination,, when tested,, then scan success rate, median read time, permission flow and recovery steps are recorded.
- Given an unapproved combination,, when proposed for event use,, then it is rejected or separately certified.

Depends on: DAVI-DISC-03, DAVI-KSK-05

DAVI-QA-04

Test degraded and offline conditions

MUST

READY

5 SP

JULY MVP

Plain English: fallback procedures are proven.

User need: As a quality engineer, I need to simulate slow, intermittent and absent connectivity.

Acceptance checks

- Given slow or lost network,, when operations run,, then the kiosk clearly reports state and does not create uncontrolled retries.
- Given queued operations,, when connectivity returns,, then they synchronize and conflicts are visible.
- Given app restart while offline,, when reopened,, then the queue remains intact.

Depends on: DAVI-OFF-04

DAVI-07 continued

Remaining user stories for this epic.

DAVI-QA-05

Conduct a full event rehearsal

MUST

BLOCKED

8 SP

JULY MVP

Plain English: staff and technology are validated together.

User need: As a event operations lead, I need to run entrance, support and freebie workflows with realistic queues.

Acceptance checks

- Given trained staff and production devices,, when rehearsal runs,, then tap, duplicate, lost card, QR fallback, manual lookup and outage scenarios are performed.
- Given rehearsal metrics,, when reviewed,, then tap latency, error rate, queue handling and manual intervention are recorded.
- Given a critical failure,, when discovered,, then the release is blocked until fixed or NFC scope is reduced.

Depends on: All July functional stories

DAVI-QA-06

Publish the event-day runbook

MUST

READY

5 SP

JULY MVP

Plain English: common issues are resolved consistently.

User need: As a operations lead, I need to give nontechnical staff a visual operating guide.

Acceptance checks

- Given the final workflow,, when documented,, then startup, station check, tap results, lost card, QR fallback, outage, override and shutdown procedures are included.
- Given each exception,, when read,, then the guide identifies the first action and escalation owner in plain language.

Depends on: DAVI-QA-05

DAVI-07 continued

Remaining user stories for this epic.

DAVI-QA-07

Establish monitoring and incident command

MUST

READY

5 SP

JULY MVP

Plain English: issues are contained quickly.

User need: As a incident commander, I need to observe station activity and coordinate technical/operational response.

Acceptance checks

- Given event day,, when operations start,, then named Axon technical, kiosk technical, entrance, freebie and support owners are available.
- Given a lost device or error spike,, when reported,, then the station can be revoked and fallback activated.
- Given a major incident,, when declared,, then one incident commander owns the operational decision.

Depends on: DAVI-STN-04

DAVI-QA-08

Complete post-event reconciliation and review

MUST

READY

5 SP

JULY MVP

Plain English: the pilot produces trustworthy outcomes and a clear next decision.

User need: As a product and operations owners, I need to close queues, reconcile stock and capture lessons.

Acceptance checks

- Given event completion,, when closing begins,, then all offline queues synchronize and kiosk credentials/caches are revoked or deleted.
- Given attendance and freebies,, when reconciled,, then duplicates, overrides, conflicts and physical-stock variance are reviewed.
- Given findings,, when retrospective completes,, then expand, iterate or retire decisions are recorded.

Depends on: DAVI-QA-05, DAVI-QA-07

Step-by-step development work to make this happen.

1

Confirm repository and device evidence

Build the DAVI kiosk, inspect dependencies, test real cards on approved phones/readers.

2

Freeze operational rules

Approve check-in eligibility, duplicate behavior, freebie rules, replacements, overrides, and fallback authority.

3

Build Axon data foundations

Add card lifecycle, assignment history, station identity, kiosk credentials, immutable check-in and freebie ledgers.

4

Create secure Axon APIs

Expose NFC check-in, freebie validation, handover confirmation, offline sync, revocation, and dashboard endpoints.

5

Adapt kiosk UX

Replace live DAVI transaction dependency, show plain-language states, and support QR/manual fallback.

6

Prove readiness

Run automated tests, concurrency tests, real-device tests, offline drills, staff rehearsal, and post-event reconciliation.

FINAL PRODUCT RULE

If the Core Team cannot prove the repository, cards, devices, duplicate prevention, and fallback in rehearsal, the correct event-day decision is QR/manual baseline - not an unproven NFC launch.